

Why This Is Master's-Worthy

A meeting pitch for a source-grounded research advisor

Prepared as a supervisor-facing explanation of academic value, technical contribution, and project impact
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Core pitch

This is not just a publication website or chatbot. It is a full-paper research intelligence platform that turns TTLAB papers into searchable evidence, citation-grounded answers, thesis extension ideas, paper intelligence artifacts, topic/author discovery, evaluation metrics, and reviewable outputs.

134

papers
local SQLite

98

extracted PDFs
full text

756

chunks
page-aware

39

topics
explorer

126

authors
profiles

Executive One-Page Pitch

This project is master's-worthy because it combines a real institutional problem, a working full-stack system, applied AI, information retrieval, human review, evaluation, and a clear research contribution. It extends an existing TTLAB publication archive concept into a source-grounded research advisor that can help students, researchers, supervisors, and the public understand and use TTLAB research.

Meeting opener

I built a platform that takes TTLAB papers beyond a publication list. It ingests papers, extracts full text, indexes evidence chunks, answers questions with citations, recommends thesis extensions, generates reviewable paper intelligence, and gives supervisors a way to evaluate and review the outputs.

Question your supervisor may ask	Short answer to give
What is the project?	A research intelligence platform for TTLAB papers: full-paper ingestion, grounded search and Q&A, thesis idea recommendation, paper intelligence, topic/author exploration, review, and evaluation.
Why is it not just a chatbot?	The chatbot is only one interface. The core work is the end-to-end pipeline: ingestion, extraction, chunking, retrieval, citations, recommendation logic, artifact generation, review workflow, and evaluation.
What makes it master's-level?	It addresses retrieval quality, source grounding, applied local LLMs, human-in-the-loop review, system evaluation, and a meaningful research advising use case.
What is the main differentiator?	The Thesis Extension Finder maps student interests, skills, timeline, difficulty, and data constraints to source-cited TTLAB papers and possible extension projects.
What is the current evidence?	Local demo data includes 134 papers, 98 extracted PDFs, 756 searchable chunks, 39 topics, 126 authors, 14 stored grounded Ask answers, and generated reviewable artifacts.

The most important message: this project is valuable because it turns archived research into action. It helps a student move from "there are many papers" to "here is a paper I can extend, here is why it fits me, here are the source citations, and here is how I could evaluate it."

1. The Problem Worth Solving

Academic publication archives are useful, but they usually stop at listing titles, authors, years, venues, and links. For a student or external stakeholder, that is not enough. They still need to know what the papers actually say, which papers relate to their interest, who works in the area, what project could be extended, and whether an AI-generated answer can be trusted.

- Students need thesis direction, not only a publication list.
- Researchers need their work to be discoverable and understandable beyond exact title searches.
- The public and industry partners need plain-language access to TTLAB research areas.

- Supervisors need review and evaluation controls before generated outputs are trusted publicly.

Value framing

The platform solves a discovery-to-advice gap. It moves from "find a paper" to "understand the paper, ask questions about it, find related work, and identify a realistic extension."

2. How This Extends The Original Paper

The original TTLAB paper automated collection, display, summarization, email notification, and manual podcast preparation. This project keeps that spirit but goes deeper: it inspects full papers, builds page-aware evidence chunks, uses retrieval and grounding, supports local LLMs, and adds thesis recommendation, review, and evaluation workflows.

Original archive direction	This project adds
Collect and display publications	Direct TTLAB discovery, seed import, SQLite storage, paper browser, paper detail, and metadata review.
Lay summaries	Public and technical paper intelligence outputs with citations, grounding status, warnings, and review status.
Manual podcast preparation	Text-only podcast scripts with cited source chunks, not ungrounded audio generation.
Future RAG support	Working Ask TTLAB with keyword/semantic/hybrid retrieval, citations, page ranges, local Ollama support, and fallback providers.
Publication ranking/display	Topic Explorer, Author Explorer, related papers, and source-derived expertise summaries.
Limited evaluation discussion	Evaluation Dashboard, test cases, coverage metrics, review templates, and admin audit events.

3. Why It Is Master's-Worthy

Dimension	Why it matters academically	Evidence in this system
Applied AI with grounding	The project deals with hallucination risk by requiring citations, source chunks, warnings, and review status.	Ask TTLAB, Paper Intelligence, Extension Finder, citation verifiers, grounding statuses.
Information retrieval	It implements and compares keyword, semantic, and hybrid retrieval over full-paper chunks.	FTS index, hashing embeddings, hybrid scoring, paper-specific Ask, diagnostics.
Human-centered research advising	It supports student decision-making by matching interests, skills, timeline, difficulty, and data constraints to papers.	Thesis Extension Finder with ranked source-cited recommendations.

Dimension	Why it matters academically	Evidence in this system
Full-stack implementation	It is not a notebook prototype; it has backend APIs, persistent models, frontend flows, CLI tooling, and docs.	FastAPI, SQLite, React/Vite, tests, demo script, reproducibility docs.
Evaluation and review	The project acknowledges quality limits and creates mechanisms to measure, review, approve, reject, and correct outputs.	Evaluation Dashboard, Admin Review, ReviewEvent audit trail, human review templates.
Local/offline AI path	It can demonstrate without cloud dependence and can benchmark local models on real hardware.	Offline deterministic providers, Ollama model registry, benchmark CLI, fallback behavior.

Master's thesis angle

A strong thesis framing is: "How can a source-grounded research intelligence platform transform an academic publication archive into a student-facing research advisor while preserving traceability, reviewability, and evaluation?"

4. What The System Does

The platform is organized as a pipeline. Each stage creates evidence that later stages use. This matters because it prevents the AI layer from becoming a black box: answers, summaries, recommendations, and scripts can point back to source chunks, pages, and papers.

Stage	How it works now	Value created
Discovery and import	TTLAB publication metadata is discovered/imported into SQLite with careful review status defaults.	Creates a reliable local paper catalogue.
PDF download and extraction	Direct PDFs are downloaded safely; PyMuPDF extracts full text; extraction status is tracked.	Moves beyond abstract-only archives.
Chunking and indexing	Full text is split into page-aware chunks and indexed for keyword and semantic retrieval.	Creates citable evidence units.
Search and Ask	Users search or ask questions over chunks; answers include citations, snippets, page ranges, and grounding status.	Makes research queryable and auditable.
Thesis Extension Finder	Student profile inputs are converted into retrieval queries and scored paper recommendations.	Turns papers into actionable project ideas.
Paper Intelligence	Summaries, methods, limitations, future work, extensions, skills, evaluation plans, and podcast scripts are generated.	Creates reusable paper-level understanding.
Explorer and review	Topics/authors/related papers support discovery; admin review and evaluation support responsibility.	Makes the demo supervisor-ready and academically defensible.

5. Current Working Evidence

These numbers come from the current local diagnostics. They show this is not just a proposed architecture; it is a working platform with real data flowing through it.

134 papers imported/discovered	98 extracted PDFs full text	756 searchable chunks page-aware
39 topics explorer	126 authors profiles	21 extension ideas stored
14 Ask answers all grounded	14 artifacts reviewable	5 podcast scripts text-only

Important caveat to say out loud

The system is demo-ready, not publication-ready. The generated outputs are source-grounded where possible, but they still need human review before public release. That limitation is not hidden; it is designed into the product through review statuses, warnings, and audit events.

6. The Main Value: Thesis Extension Finder

The Thesis Extension Finder is the strongest differentiator because it directly addresses a student need: choosing a feasible project or thesis direction from TTLAB research. It does not simply summarize papers. It separates source facts from system suggestions and explains fit, skills, risk, difficulty, data needs, and evaluation.

Output section	Why it matters
Paper focus	Explains what the source paper actually does.
Source-supported facts	Shows the cited evidence used for the recommendation.
Identified gap	Marks whether limitations/future work are explicit, inferred, or not found.
Extension idea	Clearly labels the new idea as a system suggestion, not a paper claim.
Why it fits the student	Connects interests, skills, project type, time, and data constraints.
MVP and stretch goals	Makes the idea feasible for a semester project.
Evaluation plan	Forces the recommendation to be measurable, not just creative.

Output section	Why it matters
Potential researcher fit	Uses authorship only and avoids claiming confirmed supervision.

Meeting line: "This is the part that makes the system more than a search tool. It helps a student turn TTLAB research into a realistic next project, while still showing what is cited from the paper and what is only a suggested extension."

7. Grounded AI, Not Blind Generation

The project deliberately avoids presenting AI output as truth. Retrieval happens first. The model or deterministic provider composes from a compact citation pack. Then grounding checks and warnings determine whether the result is grounded, partial, or unsupported. Local LLMs are optional and fall back safely when unavailable.

Risk	Project response
Hallucinated answer	Answers cite retrieved chunks and include grounding status. Unsupported or weak answers are marked.
Misleading summaries	Paper Intelligence stores citations, source chunk IDs, warnings, and needs_review status.
Unreviewed public output	Admin Review exists so outputs can be approved, rejected, corrected, or marked for reprocess.
Cloud dependency	Offline deterministic providers work without API keys; Ollama support is optional for local model answers.
Weak evaluation claims	Evaluation Dashboard distinguishes not_run, sample, and available results instead of inventing metrics.

8. Stakeholder Value

The project is valuable because it has more than one user. A strong master's project should not only be technically interesting; it should also solve a meaningful problem for real stakeholders.

Stakeholder	Value delivered
Student	Finds papers that match interests and gets realistic extension ideas with required skills, risk, difficulty, data constraints, MVP scope, and evaluation plan.
Supervisor	Can discuss project directions using cited source evidence, not only the student's interpretation.
Researcher	Gets paper summaries, related work, topic visibility, author profiles, and public-friendly outputs.
TTLAB/public visitor	Can ask natural questions, browse topics/authors, and understand research without reading every PDF first.
Administrator/reviewer	Can correct metadata, approve/reject generated outputs, and keep an audit trail.

9. Demo Narrative For The Meeting

Step	Screen	What to emphasize
1	Dashboard	Show papers, PDFs, chunks, topics, authors, artifacts, review queue, and evaluation status.
2	Papers and detail	Open a paper and show metadata, extraction status, chunks, related papers, and paper intelligence.
3	Search	Search RAG, agriculture, AI, optimization, or another area and show chunk-level results.
4	Ask TTLAB	Ask a natural question and point to citations, page ranges, and grounding status.
5	Thesis Extension Finder	Enter interests and skills; show extension ideas, MVP scope, risk, evaluation, and citations.
6	Topic/Author Explorer	Show topics, authors, source-derived expertise, and related work.
7	Admin and Evaluation	Show review controls and explain that generated outputs are not treated as final.

Best meeting question to ask the supervisor

"Which 5 to 10 TTLAB papers should become the first manually reviewed demo set, and what criteria would you use to judge whether a thesis extension recommendation is useful?"

10. Research Questions And Evaluation Plan

The thesis can be defended around specific evaluation questions rather than vague "AI helps research" claims. The platform already has the hooks needed to evaluate retrieval, answer grounding, recommendation usefulness, artifact faithfulness, and human review outcomes.

Research question	Possible evidence
Can full-paper retrieval improve discovery over metadata-only browsing?	Retrieval test questions, Recall@k, MRR, cited paper coverage, qualitative examples.
Can citation-grounded answers reduce unsupported claims?	Grounding status counts, citation correctness review, unsupported-claim warnings.
Can the Thesis Extension Finder help students identify feasible projects?	Supervisor/student review scores for relevance, feasibility, usefulness, and risk appropriateness.
Are generated paper artifacts faithful enough to support review workflows?	Artifact citation coverage, faithfulness ratings, reviewer corrections, needs_review/approved rates.
How do local LLMs affect speed and answer quality?	Ollama benchmark latency, tokens/sec, citation compliance, answer quality ratings.

11. Limitations That Make The Project Honest

A supervisor will likely respect the project more if limitations are clear. The current system is designed as a local/demo MVP and it intentionally does not overclaim production readiness.

- Generated outputs are AI-assisted drafts and require review before public use.
- Some PDFs are missing direct links, and scanned/image-heavy PDFs may need OCR later.
- Topic and author expertise labels are deterministic and inferred unless reviewed.
- Evaluation files are scaffolds unless manually reviewed gold sets are added.
- No authentication, production deployment, or public TTLAB website integration is included yet.
- Local Ollama depends on the user's machine and service availability; offline fallback remains available.

Why this is still a strength

These limitations are exactly where a thesis can be rigorous: evaluation, human review, source grounding, and responsible deployment boundaries become part of the academic contribution.

12. Suggested Closing Pitch

Use this as the final 30 to 45 seconds of the meeting:

"The reason I think this is master's-worthy is that it is not only applying an LLM to papers. It is building a full research intelligence pipeline around TTLAB: ingesting papers, extracting full text, retrieving source evidence, answering with citations, helping students find thesis extensions, generating reviewable paper outputs, mapping topics and authors, and measuring quality through evaluation and review. It directly extends the original publication archive idea into something more useful for students, supervisors, researchers, and the public. The next best step is to choose a small reviewed paper set and evaluate whether the recommendations and grounded answers are useful enough for a real TTLAB demo."

Appendix A: Concrete Talking Points

If he asks...	Say...
What is the academic novelty?	The value is in the integrated, source-grounded research advisor: full-paper retrieval, citation-grounded generation, thesis recommendation, review, and evaluation for a real institutional archive.
Why not just use ChatGPT with PDFs?	This system is persistent, searchable, auditable, institution-specific, citation-aware, reviewable, and built around TTLAB's paper corpus and thesis/project workflows.
Why does this matter to TTLAB?	It makes TTLAB research easier to discover, explain, reuse, and connect to student projects or public/industry interests.
What needs supervisor input?	A manually reviewed paper subset, gold evaluation questions, criteria for good thesis recommendations, and guidance on public-facing wording.
What would be the final deliverable?	A demoable platform, evaluated pipeline, documented methodology, reviewed sample outputs, and thesis write-up evidence.

Appendix B: Useful Demo Questions

- Which papers discuss RAG or retrieval augmented generation?
- What research relates to agriculture or AI?
- Which paper could I extend if I know Python, React, and FastAPI?
- Who appears to work on optimization or machine learning?
- What are the limitations and future work for this paper?

Appendix C: Current Evidence Snapshot

Area	Current evidence
Dataset	134 papers available locally; 98 extracted PDFs; 35 missing direct PDFs.
Retrieval	756 searchable chunks; 756 keyword-indexed chunks; 25 semantic-indexed chunks using hashing embeddings.
Ask TTLAB	14 stored Ask answers, all currently marked grounded in diagnostics.
Thesis Extension Finder	5 recommendation runs and 21 generated recommendations, currently partial because some future-work support is inferred rather than explicit.
Paper Intelligence	14 artifacts across 5 papers, including 5 text-only podcast scripts.
Explorer	39 topics, 126 authors, 767 paper-topic links, and 1105 author-topic links.
Review and evaluation	167 items in local review queue; extension and artifact evaluation results available; retrieval and QA result files not currently run.